

* $\{a_n\} = \left\{ \left(\frac{n}{n+1} \right)^{n^2+1} \right\}$ dizisinin yakınsaklığını inceleyin.

$$\lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^{n^2+1} = \lim_{n \rightarrow \infty} \left[\underbrace{\left(1 - \frac{1}{n+1} \right)^{n+1}}_{e^{-1}} \right]^{\frac{n^2+1}{n+1}} = e^{-\infty} = 0 \Rightarrow \text{dizi yakınsak}$$

Soru 1. Genel terimi $a_n = n - \frac{1}{2} \ln(1 + e^{2n})$, ($n=1,2,\dots$), olan dizinin limitini bulunuz.

Cevap 1.

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \left[n - \frac{1}{2} \ln(1 + e^{2n}) \right]$$

$$(\Theta 5) = \lim_{n \rightarrow \infty} \left[\ln(e^n) - \ln(1 + e^{2n})^{1/2} \right]$$

$$(\Theta 5) = \lim_{n \rightarrow \infty} \ln \left(\frac{e^n}{\sqrt{1 + e^{2n}}} \right)$$

$$(\Theta 5) = \lim_{n \rightarrow \infty} \ln \left(\frac{1}{\sqrt{\frac{1}{e^{2n}} + 1}} \right)$$

$$(\Theta 5) = \ln \left(\lim_{n \rightarrow \infty} \frac{1}{\sqrt{\frac{1}{e^{2n}} + 1}} \right) = \ln \left(\frac{1}{\lim_{n \rightarrow \infty} \sqrt{\frac{1}{e^{2n}} + 1}} \right)$$

$$(\Theta 5) = \ln \left(\frac{1}{\sqrt{0+1}} \right) = \ln(1)$$

$$= 0$$

Soru 3. a) $\sum_{n=0}^{\infty} \frac{\pi^{-n}}{\cos(n\pi)}$ serisinin toplamını bulunuz. (10 puan)

$$\sum_{n=0}^{\infty} \frac{\pi^{-n}}{\cos(n\pi)} = 1 - \frac{1}{\pi} + \frac{1}{\pi^2} - \frac{1}{\pi^3} + \dots + (-1)^n \pi^{-n} + \dots = \sum_{n=0}^{\infty} \left(-\frac{1}{\pi}\right)^n \quad (2)$$

geometrik seridir. Burada $a=1$ $r=-\frac{1}{\pi}$ dir.

(4) (2) $|r| < 1$ olduğundan serinin toplamı

$$S = \frac{a}{1-r} = \frac{1}{1-(-\frac{1}{\pi})} = \frac{\pi}{\pi+1} \quad (2)$$

b) Genel terimi $a_n = \left(\frac{3n-1}{3n+2}\right)^n$ olan $\{a_n\}$ dizisinin limitini bulunuz.

$$\begin{aligned} \lim_{n \rightarrow \infty} a_n &= \lim_{n \rightarrow \infty} \left(1 - \frac{3}{3n+2}\right)^n = \lim_{n \rightarrow \infty} \left[\left(1 - \frac{3}{3n+2}\right)^{3n+2-2} \right]^{\frac{1}{3}} \\ &= \lim_{n \rightarrow \infty} \left[\left(1 - \frac{3}{3n+2}\right)^{3n+2} \cdot \left(1 - \frac{3}{3n+2}\right)^{-2} \right]^{\frac{1}{3}} \end{aligned}$$

(25) 5,232323... sayısının serileri kullanarak iki tamsayının oranı olarak ifade ediniz.

$$5,232323 \dots = 5 + \frac{23}{100} + \frac{23}{(100)^2} + \frac{23}{(100)^3} + \dots$$

$$= 5 + \sum_{n=1}^{\infty} \frac{23}{100} \left(\frac{1}{100}\right)^{n-1} \quad a = \frac{23}{100} \quad r = \frac{1}{100}$$

$$|r| = \frac{1}{100} < 1 \rightarrow \sum_{n=1}^{\infty} \frac{23}{100} \left(\frac{1}{100}\right)^{n-1} = \frac{a}{1-r} = \frac{\frac{23}{100}}{1-\frac{1}{100}} = \frac{23}{99}$$

(Ser. yalınlaşık)

$$5,232323 \dots = 5 + \frac{23}{99} = \frac{518}{99}$$

$$*) \sum_{n=1}^{\infty} \frac{4}{n^2+4n+3} = ?$$

$$\frac{4}{n^2+4n+3} = \frac{A}{n+3} + \frac{B}{n+1} \quad \Rightarrow \boxed{A=-2 \mid B=2}$$

$$\sum_{n=1}^{\infty} \frac{4}{n^2+4n+3} = 2 \sum_{n=1}^{\infty} \frac{1}{n+1} - \frac{1}{n+3}$$

$$S_n = 2 \left[\left(\frac{1}{2} - \frac{1}{4} \right) + \left(\frac{1}{3} - \frac{1}{5} \right) + \left(\frac{1}{4} - \frac{1}{6} \right) + \dots + \left(\frac{1}{n} - \frac{1}{n+2} \right) + \left(\frac{1}{n+1} - \frac{1}{n+3} \right) \right]$$

$$S_n = 2 \left[\frac{5}{6} - \frac{1}{n+2} - \frac{1}{n+3} \right] \Rightarrow \lim_{n \rightarrow \infty} S_n = \frac{5}{3} \Rightarrow \sum_{n=1}^{\infty} \frac{4}{n^2+4n+3} = \frac{5}{3}$$

$$*) \sum_{k=2}^{\infty} \ln\left(\frac{k-1}{k}\right) \quad \text{serisinin toplamını bulup sonucu yorumlayın.}$$

$$S_n = \ln \frac{1}{2} + \ln \frac{2}{3} + \ln \frac{3}{4} + \dots + \ln \frac{n-1}{n}$$

$$= \ln \left(\frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \cdot \dots \cdot \frac{n-1}{n} \right) = \ln \left(\frac{1}{n} \right) \Rightarrow \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \ln \left(\frac{1}{n} \right) = -\infty$$

$\Downarrow$

Seri $-\infty$ 'a
iraksar.

$$(*) \sum_{n=1}^{\infty} \frac{3^{n-1}-1}{6^{n-1}} = ?$$

$$\sum_{n=1}^{\infty} \frac{3^{n-1}-1}{6^{n-1}} = \sum_{n=1}^{\infty} \frac{1}{2^{n-1}} - \sum_{n=1}^{\infty} \frac{1}{6^{n-1}} = 2 - \frac{6}{5} = \frac{4}{5}$$

$$a=1 \quad r=\frac{1}{2}$$

$$|r| = \frac{1}{2} < 1$$

$$\text{Toplam} = \frac{a}{1-r}$$

$$= \frac{1}{1-\frac{1}{2}} = 2$$

$$a=1 \quad r=\frac{1}{6}$$

$$|r| = \frac{1}{6} < 1$$

$$\text{Toplam} = \frac{a}{1-r} = \frac{1}{1-\frac{1}{6}}$$

$$= \frac{6}{5}$$

⑧ $\sum_{n=1}^{\infty} \ln \sqrt{n+1} - \ln \sqrt{n}$ serisinin n . kısmi toplamı için bir formül bulunuz ve bu formül yardımıyla serinin yakınsaklığını inceleyiniz.

$$\sum_{n=1}^{\infty} -\ln \sqrt{n} + \ln \sqrt{n+1} \Rightarrow S_n = \cancel{-\ln 1} + \cancel{\ln 2} - \cancel{\ln 2} + \cancel{\ln 3} - \dots - \cancel{\ln n} + \ln \sqrt{n+1} \\ = \ln \sqrt{n+1}$$

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \ln \sqrt{n+1} = +\infty \Rightarrow \text{Seri } +\infty \text{ 'e ıraksar}$$

⑨ $\sum_{n=1}^{\infty} \arccos \frac{1}{n+1} - \arccos \frac{1}{n+2}$ serisinin n . kısmi toplamı için bir formül bulup yakınsaklığını inceleyiniz. Yakınsak ise değerini bulunuz.

$$S_n = \arccos \frac{1}{2} - \cancel{\arccos \frac{1}{3}} + \cancel{\arccos \frac{1}{3}} - \cancel{\arccos \frac{1}{4}} + \dots + \cancel{\arccos \frac{1}{n+1}} - \arccos \frac{1}{n+2}$$

$$S_n = \underbrace{\arccos \frac{1}{2}}_{\pi/3} - \arccos \frac{1}{n+2} = \frac{\pi}{3} - \arccos \frac{1}{n+2}$$

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \frac{\pi}{3} - \underbrace{\arccos \frac{1}{n+2}}_{\pi/2} = \frac{\pi}{3} - \frac{\pi}{2} = -\frac{\pi}{6} \rightarrow \text{Seri yakınsaktır.} \\ \text{Toplamı } -\frac{\pi}{6} \text{ dir.}$$

⑩ $\{a_n\} = \left\{ \left(1 + \frac{1}{n^2}\right)^n \right\}$ dizisinin yakınsaklığını inceleyin.

I. Yol

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n^2}\right)^n = \lim_{n \rightarrow \infty} \left[\underbrace{\left(1 + \frac{1}{n^2}\right)^{n^2}}_e \right]^{1/n \rightarrow 0} = e^0 = 1 \Rightarrow \text{Dizi yakınsaktır.}$$

II. Yol Logaritmik limit ile de çözülebilir.

$$6) \sum_{n=1}^{\infty} \frac{1}{n(n+2)} = ?$$

$$\frac{1}{n(n+2)} = \frac{A}{n} + \frac{B}{n+2} \quad A = \frac{1}{2} \quad B = -\frac{1}{2}$$

$$\sum_{n=1}^{\infty} \frac{1}{n(n+2)} = \frac{1}{2} \sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+2} \right)$$

$$S_n = \frac{1}{2} \left[\left(1 - \frac{1}{3}\right) + \left(\frac{1}{2} - \frac{1}{4}\right) + \left(\frac{1}{3} - \frac{1}{5}\right) + \dots + \left(\frac{1}{n-2} - \frac{1}{n}\right) + \left(\frac{1}{n-1} - \frac{1}{n+1}\right) + \left(\frac{1}{n} - \frac{1}{n+2}\right) \right]$$

$$= \frac{1}{2} \left[\frac{3}{2} - \frac{1}{n+1} - \frac{1}{n+2} \right] \Rightarrow \lim_{n \rightarrow \infty} S_n = \frac{1}{2} \left(\frac{3}{2} - \frac{1}{n+1} - \frac{1}{n+2} \right) = \frac{3}{4}$$

$$\Downarrow$$

$$\sum_{n=1}^{\infty} \frac{1}{n(n+2)} = \frac{3}{4}$$

$$*) \sum_{n=1}^{\infty} \frac{3n^2+3n+1}{n^3(n+1)^3} \quad \text{serisinin toplamını bulunuz.}$$

$$\frac{3n^2+3n+1}{n^3(n+1)^3} = \frac{(1+n)^3 - n^3}{n^3(1+n)^3} = \frac{1}{n^3} - \frac{1}{(n+1)^3} \quad \text{olduğundan}$$

$$\sum_{n=1}^{\infty} \frac{3n^2+3n+1}{n^3(n+1)^3} = \sum_{n=1}^{\infty} \left(\frac{1}{n^3} - \frac{1}{(n+1)^3} \right) \quad \text{dizisi}$$

$$S_n = \left(1 - \frac{1}{2^3}\right) + \left(\frac{1}{2^3} - \frac{1}{3^3}\right) + \left(\frac{1}{3^3} - \frac{1}{4^3}\right) + \dots + \left(\frac{1}{(n-1)^3} - \frac{1}{n^3}\right) + \left(\frac{1}{n^3} - \frac{1}{(n+1)^3}\right)$$

$$= 1 - \frac{1}{(n+1)^3}$$

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{(n+1)^3} \right) = 1 \Rightarrow \sum_{n=1}^{\infty} \frac{3n^2+3n+1}{n^3(n+1)^3} = 1$$

$$\textcircled{*} \frac{2}{3!} + \frac{3}{4!} + \frac{4}{5!} + \dots = ?$$

$$\frac{2}{3!} + \frac{3}{4!} + \frac{4}{5!} + \dots = \sum_{n=1}^{\infty} \frac{n+1}{(n+2)!}$$

$$\frac{n+1}{(n+2)!} = \frac{n+2-1}{(n+2)!} = \frac{n+2}{(n+2)!} - \frac{1}{(n+2)!} = \frac{1}{(n+1)!} - \frac{1}{(n+2)!}$$

olduğundan

$$\sum_{n=1}^{\infty} \frac{n+1}{(n+2)!} = \sum_{n=1}^{\infty} \left(\frac{1}{(n+1)!} - \frac{1}{(n+2)!} \right) \text{ dir.}$$

$$S_n = \left(\frac{1}{2!} - \frac{1}{3!} \right) + \left(\frac{1}{3!} - \frac{1}{4!} \right) + \dots + \left(\frac{1}{(n+1)!} - \frac{1}{(n+2)!} \right) = \frac{1}{2!} - \frac{1}{(n+2)!}$$

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left(\frac{1}{2} - \frac{1}{(n+2)!} \right) = \frac{1}{2} \Rightarrow \frac{2}{3!} + \frac{3}{4!} + \dots = \underline{\underline{\frac{1}{2}}}$$

$\textcircled{*} X = 2,131313\dots = 2,\overline{13}$ sayısını serileri kullanarak iki tamsayının oranı olarak ifade ediniz.

$$x = 2,\overline{13} = 2 + \underbrace{\frac{13}{100} + \frac{13}{(100)^2} + \frac{13}{(100)^3} + \dots}_{\text{Geometrik Seri}} = 2 + \sum_{n=1}^{\infty} \frac{13}{100} \cdot \left(\frac{1}{100} \right)^{n-1}$$

$$a = \frac{13}{100} \quad r = \frac{1}{100}$$

$$|r| = \frac{1}{100} < 1 \Rightarrow \sum_{n=1}^{\infty} \frac{13}{100} \cdot \left(\frac{1}{100} \right)^{n-1} = \frac{a}{1-r} = \frac{\frac{13}{100}}{1 - \frac{1}{100}} = \frac{13}{99}$$

seri yakınsaktır

$$x = 2 + \frac{13}{99} = \frac{211}{99}$$

$$\textcircled{*} 4 - 1 + \frac{1}{4} - \frac{1}{16} + \frac{1}{64} - \dots = ? \Rightarrow a = 4 \quad r = -\frac{1}{4} \text{ Geometrik Seridir}$$

$\xrightarrow{-1/4} \xrightarrow{-1/4} \xrightarrow{-1/4} \xrightarrow{-1/4}$

$$\sum_{n=1}^{\infty} 4 \cdot \left(-\frac{1}{4} \right)^{n-1} \Rightarrow |r| = \frac{1}{4} < 1 \text{ Seri } \frac{a}{1-r} \text{ 'ye yakınsar.}$$

$$\frac{a}{1-r} = \frac{4}{1 - (-\frac{1}{4})} = \frac{16}{5} \Rightarrow 4 - 1 + \frac{1}{4} - \frac{1}{16} - \dots = \underline{\underline{\frac{16}{5}}}$$

(7)